Systematic reviews and PRISMA

Workflow lab: Goal → Approach → Execution → Check → Report.

Learning objectives

- Frame a systematic-review question using PICO.
- Draft a reproducible search strategy across at least two databases.
- Produce a PRISMA flow diagram from counts at each screening stage.

Prerequisites

None beyond reading a paper.

Background

A systematic review is itself a study: it has a protocol, a pre-specified search strategy, explicit inclusion and exclusion criteria, and a plan for synthesis. The PRISMA 2020 statement is the reporting standard, which means the diagram showing what you searched, what you excluded, and why is not optional. PROSPERO (https://www.crd.york.ac.uk/prospero/) is the standard registry for review protocols — registering early protects you from post-hoc drift.

Good search strategies combine controlled-vocabulary terms (MeSH on PubMed, Emtree on Embase) with free-text terms, connect concepts with Boolean AND, and expand synonyms with OR. A librarian-reviewed strategy typically outperforms a DIY one by a wide margin in recall; co-authoring with a health-sciences librarian is the single best investment in review quality.

Setup

```r
library(tidyverse)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Illustrate a PICO-framed question, sketch a search strategy, and simulate screening counts to generate a PRISMA flow diagram.

2. Approach

A fictional question in PICO form:

> In adults with type 2 diabetes (**P**), does metformin (**I**) compared with placebo (**C**) reduce all-cause mortality (**O**) at 24 months?

A sketch of a search strategy (PubMed syntax):

```
("Diabetes Mellitus, Type 2"[MeSH] OR "type 2 diabetes"[tiab])
AND (metformin[MeSH] OR metformin[tiab])
AND (randomised[tiab] OR randomized[tiab] OR trial[tiab] OR RCT[tiab])
```

3. Execution — simulated screening counts

```r
  stage = c("Identified (PubMed)", "Identified (Embase)",
            "After deduplication",
            "Title/abstract screened", "Full-text assessed",
            "Included in qualitative synthesis",
            "Included in meta-analysis"),
  n = c(812, 640, 1098, 1098, 84, 22, 18)
)
prisma
```

```r
prisma |>
  mutate(stage = factor(stage, levels = rev(stage))) |>
  ggplot(aes(n, stage)) +
  geom_col(fill = "#1a73e8", alpha = 0.8) +
  geom_text(aes(label = n), hjust = -0.1, size = 3.5) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(x = "n", y = NULL)
```

4. Check

PRISMA 2020 checklist pointer (items most often missed):

- Item 5: eligibility criteria pre-specified.
- Item 6: information sources with date of last search.
- Item 7: complete search strategy for every database, including limits.
- Item 16a: reasons for excluding full-text articles.

5. Report

> A systematic review of the effect of metformin on all-cause mortality in adults with type 2 diabetes was conducted following PRISMA 2020. Database searches identified 812 records from PubMed and 640 from Embase. After deduplication, 1098 records were screened by title and abstract; 84 full texts were assessed, and 22 studies were included in the qualitative synthesis, of which 18 were pooled by meta-analysis.

Common pitfalls

- Ad-hoc searches that cannot be re-run.
- Failing to double-screen at each stage (reviewer drift).
- Reporting included-study counts without the flow that produced them.

Further reading

- Page MJ et al. (2021). *The PRISMA 2020 statement.* BMJ.
- Cochrane Handbook for Systematic Reviews of Interventions.

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)